Branden Palmer

Active Top Secret/SCI

205-353-1613 / brandenpalmer08@gmail.com / github.com/palmerusaf

Technical Skills

- **Languages:** Java, TypeScript, JavaScript, Python, SQL
- **Frameworks & Databases:** React, JavaFX, PostgreSQL, SQLite, PGlite, Drizzle ORM
- **Tools & Practices:** Git, GitHub Actions, CI/CD, regression testing, Linux

Education

University of Texas at Austin | *Austin, TX, USA* **Jan 2026 – May 2029**

- M.S. in Computer Science (in progress) | GPA: 4.00

Arizona State University | *Tempe, AZ, USA* **Aug 2020 – Dec 2025**

- B.S. in Computer Science | **Summa Cum Laude** | GPA: 3.83

Software Engineering & Open Source

Plotly.js | *Link* | *Open-source JavaScript visualization library* **Dec 2025 – Aug 2026**

- Co-developed **map camera auto-centering and fit-to-data** functionality in the Plotly.js core library, including geographic bounds handling and **anti-meridian wrapping**; included by default in v4.
- **Optimized and benchmarked performance** for large point datasets and added targeted **regression tests** across map configurations.
- Delivered a core-library feature that also benefits downstream integrations, including **Plotly.py** (~48M monthly downloads at the time measured).

G8 Medical Technology | *Code* | *Team-developed medical application* **Jan 2024 – Apr 2024**

Java JavaFX SQLite

- Collaborated on a **Java/JavaFX application backed by SQLite**; organized a six-person team into three feature-focused groups to coordinate implementation and deliver requirements on time.
- Integrated team contributions through **37 Git merges**, supporting development across the shared codebase.

Super Productivity | *Link* | *Open-source productivity application* **Apr 2022 – Jan 2024**

SCSS Angular.js

- Used **git bisect** to identify a regression, isolated the problematic **SCSS**, and contributed a fix merged into the production branch.

Projects

Job Sourcerer | *Demo* | *Code* | *Job application management platform* **Jan 2025 – Present**

TypeScript React PostgreSQL PGlite Drizzle ORM GitHub Actions SBERT

- Built a **browser extension architecture** using background, content, and popup scripts to collect job postings and persist structured records locally.
- Implemented a **client-side data layer** with Drizzle ORM and PGlite for job, resume, and analytics data without a backend service.
- Set up **GitHub Actions CI/CD** to detect regressions and streamline deployment to Chrome and Firefox extension stores; implemented resume matching using **SBERT and cosine similarity**.

Autonomous Driving Trajectory Prediction | *University of Texas at Austin* | *Deep learning project*

Python PyTorch Transformers CNNs NumPy **Jan 2026 – Apr 2026**

- Developed and trained **PyTorch models** to predict vehicle trajectories using road geometry and image inputs.
- Built reusable **training and evaluation pipelines** with data loading, optimization, checkpointing, and quantitative trajectory-error evaluation.

Applied Stylometry | *Report* | *Forensic Text Analysis* | *Historical FBI Case Study* **Nov 2025 – Dec 2025**

Python BeautifulSoup pandas scikit-learn NLP TF-IDF PCA Plotly

- Developed a **Python text-analysis pipeline** to extract and structure publicly available documents from the historical FBI Unabomber investigation using **BeautifulSoup and pandas**.
- Applied **TF-IDF, character n-grams, PCA, and stylometric analysis** to investigate linguistic similarities across document collections.
- Built **interactive Plotly visualizations** to explore document clustering and evaluated the reliability and limitations of automated authorship analysis.

Military Experience

United States Air Force | *F-16 Crew Chief* | *Prior active-duty service*

- Served **seven years on active duty** in F-16 aircraft maintenance, gaining experience in mission-critical operations, technical troubleshooting, and team coordination.